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Conservation biology for the commercial insect trade in Japan: agricultural bumblebees and companion insects as examples

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Summary

Japan imports a wide range of arthropods for industrial use and as companion animals. Such imports may threaten ecosystems locally and in their regions of origin. Two iconic insect imports that pose ecological problems are agricultural bumblebees and companion beetles.

Colonies of the bumblebee *Bombus terrestris* have contributed significantly to agricultural production since they were first brought to Japan in the 1990s. But, in their progressive feralisation, they harm populations of native bumblebees through competition, hybridisation, and the introduction of parasites. They also threaten native plant reproduction. The species is currently permitted for agricultural use only in netted greenhouses.

Since 2000, imports of companion beetles have thrived, with an estimated market size of many billions of yen. The popularity of rare species has led to a sharp rise in prices, overhunting, and smuggling from their native countries. These exotic species may also become invasive if they escape into nature. There are no clear restrictions on beetle imports, but a government campaign is aimed to improve ethical standards for breeding.

In addition, imported tarantulas, centipedes and scorpions are becoming increasingly popular. These species pose similar threats as imported beetles and bees, but the actual state of importation and breeding is difficult to ascertain. Importing insects into Japan can create the following issues: the overexploitation of rare species collected from their native habitats; the traffic in species of which collection and sale is prohibited; the risk that escaped individuals will breed as invasive species; and the introduction of alien microorganisms and parasites.

Keywords

Alien species – Beetle – *Bombus terrestris* – Bumblebee – Companion animal – Ecological risk – Endangered species – Pollinator.

Introduction

Insects and other arthropods are traded internationally for a variety of purposes. One is industrial use: insects can serve as natural enemies for biocontrol, as agricultural pollinators, and as animal feed. Insects are also traded as companion animals. This article focuses on the risks posed by these trades both before and after transport (as described in the introduction of this issue [1]). Serving as examples are two general imports into Japan: bees used as agricultural pollinators, and companion beetles and other arthropods. The author introduces these cases and the measures taken by the Japanese government on future international trade of insects.

The commercial use of pollinators in agriculture has a long history [2]. In recent years, bumblebees have become increasingly commercialised for this purpose. When Velthuis and van Doorn [3] reviewed the industry, they found that, in 2004, around one million bumblebee colonies were sold annually worldwide. Sales have continued to expand since then [4]. The supply of bumblebee colonies is essential for the global production of tomatoes and other horticultural products. Exact revenues from bumblebee colony sales are difficult to estimate because the companies dealing in them are private. In Japan, for example, bumblebee colonies are sold at prices between US$ 200 and US$ 300 per nest. The industry as a whole is estimated to be worth hundreds of millions of dollars.

As for the market for companion insects, the culture of raising insects has existed for a long time, and the market is expected to continue to grow.

International trade in insects can be profitable and often serves useful purposes, but it also represents the deliberate introduction of alien species into a new ecosystem. As such, it brings a variety of environmental and ecological risks. It is important to accumulate scientific analyses of such risks in order to conserve biodiversity.

Commercialisation of the European bumblebee, *Bombus terrestris*

The European bumblebee, *Bombus terrestris*, is one of the most successful biological agents for commercial pollination (Figure 1). In Japan, use of the bumblebee, mainly for pollination of tomatoes, has increased since it was introduced in 1991. The number of commodity colonies in circulation reached 70,000 in 2004 [5], a number that has remained roughly the same since then [6].

Because of their popularity, these biological agents risk becoming an invasive alien species. Since 1996, when a naturalised colony of *B. terrestris* was found in Hokkaido [7], the number of captive colonies that have escaped from glass or vinyl houses into the field has continued to increase. This means that the rate of invasion is increasing [8]. The possible ecological impacts of naturalising *B. terrestris* have sparked controversy in many countries [9, 10]. In Japan, ecologists and entomologists have warned of its ecological risks [7, 11, 12].

Risk and impact assessment of *Bombus terrestris*

The National Institute for Environmental Studies (NIES) has examined the ecological risks of alien bumblebees in Japan in collaboration with other institutes, universities, private companies, and the government and found four ecological risks posed by *B. terrestris*.

Figure 1

***Bombus terrestris* visiting *Trifolium repens* (white clover) flowers**

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- **Competitive exclusion of native bumblebees**
  The NIES field survey in Hokkaido showed that, in terms of niche overlap for limited nest sites, *Bombus hypocrita sapporoensis* and *Bombus diversus tersatus* were the two native species most likely to be affected by *B. terrestris* through competition for nest sites [13, 14]. Data collected on the distribution of *B. terrestris* in Hokkaido since 2003 indicate that the alien bee has predominantly invaded human-modified areas such as agricultural land and residential zones. It has also affected natural habitats such as coastal and alpine grasslands [15]. From this information, the author has concluded that *B. terrestris* has a high potential to push out native bumblebees in the field.

  Establishment of *B. terrestris* has been confirmed so far only in Hokkaido. No natural nests of the alien species have ever been reported in the islands to the south: Honshu, Shikoku, and Kyushu. Although the reasons for this difference in establishment are not clear, it is possible that in southern Honshu the community structure and flowering time of plants do not provide suitable food resources for the alien bumblebees, or that there are too few abandoned mammal nests for the bees to use as nesting sites.

- **Inhibition of native plant reproduction**
  The NIES found that *B. terrestris* reduces the success of pollination of the Japanese endemic flower *Corydalis ambigua* in the field. Because the flowers of *C. ambigua* have long spurs, they are usually pollinated by *Bombus ardens*, which has a long proboscis. *Bombus terrestris* has a short proboscis and cannot reach the nectar at the bottom of the spurs; it therefore steals the nectar by gnawing a hole in the side of the flower. This damage to the flower inhibits normal pollination and decreases its ability to reproduce. A field survey comparing the flowers visited by native bumblebees with those visited by *B. terrestris* showed that seed reproduction rates were significantly lower in the latter [16].

- **Reproductive interference by crosses**
  The NIES verified that *B. terrestris* in the field interferes with the reproduction of native bumblebees through crossing. Laboratory experiments showed that crosses between native queens and *B. terrestris* males produced non-viable hybrid eggs; for all practical purposes, these native queens were rendered sterile [17]. Before the start of spring nesting, NIES researchers caught native *B. hypocrita* queens on

Hokkaido and in the Tohoku district of Honshu and analysed the DNA sequences of the spermatozoa stored in their spermathecae. The analyses showed that about 30% of the queens had mated with *B. terrestris* males, likely escaped from greenhouses (Figure 2). Given that a *Bombus* queen generally mates only once in her life, these high frequencies of interspecific mating with *B. terrestris* pose serious threats to the populations of native bumblebees in Japan [18].

– **Alien parasites**
Alien parasites occasionally cause serious damage to native species. For example, the varroa mite, *Varroa destructor*, is native to eastern Asia where it parasitises the eastern honeybee, *Apis cerana*, and is a virulent parasite of the European honeybee, *Apis mellifera* [19]. The mites were brought from Japan and eastern Russia and introduced into the Western Hemisphere through the transportation of honeybee colonies. The mite is now responsible for enormous losses of European bee colonies worldwide. There are more than 100 known parasite species associated with bumblebees, including viruses, fungi, protozoa, nematodes, insects and mites [20]. It is possible that many of these parasites have already been carried into Japan in imported bumblebee colonies.

As expected, in 1999 the NIES found an endoparasitic mite, *Locustacarus buchneri*, in colonies of *B. terrestris* brought to Japan [21]. In colonies imported from the Netherlands and Belgium, the average infestation rate was 20%. The NIES later investigated whether this tracheal mite had infested native bumblebee species and commercial colonies that had been collected and introduced between 1997 and 2001. The NIES detected the mites in natural populations of two species, *B. hypocrita sapporoensis* and *B. hypocrita hypocrita*. Approximately 5% of individuals were infected, and approximately 20% of commercial colonies. Because both species were once exported from Japan to Europe for commercialisation, these results indicate that bumblebee commercialisation caused overseas migration and cross-infestation with parasitic mites among natural and commercial colonies [22, 23].

The ecological and physiological burden of tracheal mites on bumblebees is not yet experimentally confirmed (K. Tsuchida, personal communication). On a positive note, no cases of *Nosema* protozoa, which has been reported to cause disease to bumblebees in other countries, have been reported to infect native populations in Japan.

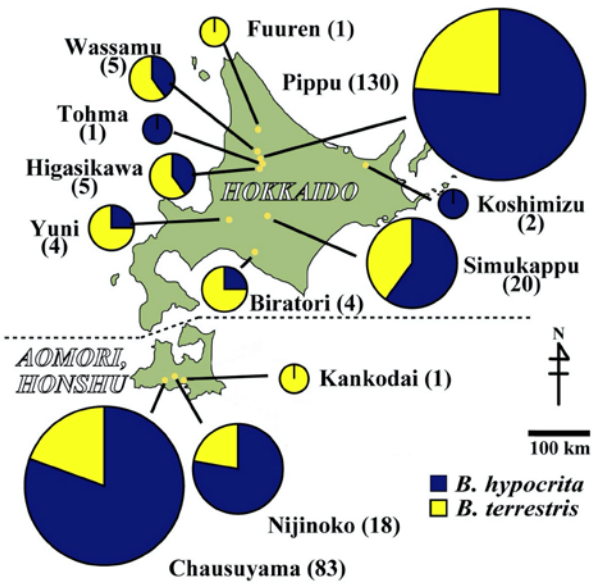

Figure 2
Paternity of spermatozoa stored in the spermatheca of native *Bombus hypocrita* queens, analysed using nuclear DNA sequences
Parentheses contain the sample size [18]

Control of invasive alien bumblebees

It has been scientifically concluded that the imported bumblebee is an invasive species with various harmful effects on native species. In response, the Ministry of the Environment of Japan has decided to regulate this species. It is difficult to ban *B. terrestris* completely because it is an essential agent for agricultural production. The government decided to allow this species to be brought to Japan for agricultural use only. Farmers can use this species in greenhouses with netting to prevent escape [24] (Figure 3).

The NIES is developing a method to eradicate *B. terrestris* in the wild using insect growth regulators and investigating a chemical treatment for *B. terrestris* using its characteristics as a social insect. Workers sprayed with pesticides while foraging in the field return to the hive and expose the larvae and queen to the pesticide, thereby inhibiting the production of a new queen [25].

The Ministry of Agriculture, Forestry and Fisheries of Japan recommends the alternative use of the native bumblebee species *Bombus ignitus* to pollinate tomato plants. The natural distribution of *B. ignitus* is limited to the islands of Honshu and Kyushu; it does not inhabit Hokkaido, where *B. terrestris* is most prevalently used and the need for an alternative pollinator is most urgent. But even a native species risks becoming invasive when transported beyond its natural habitat. The first potent impact could be genetic introgression caused by crosses between natural and commercial colonies. Through

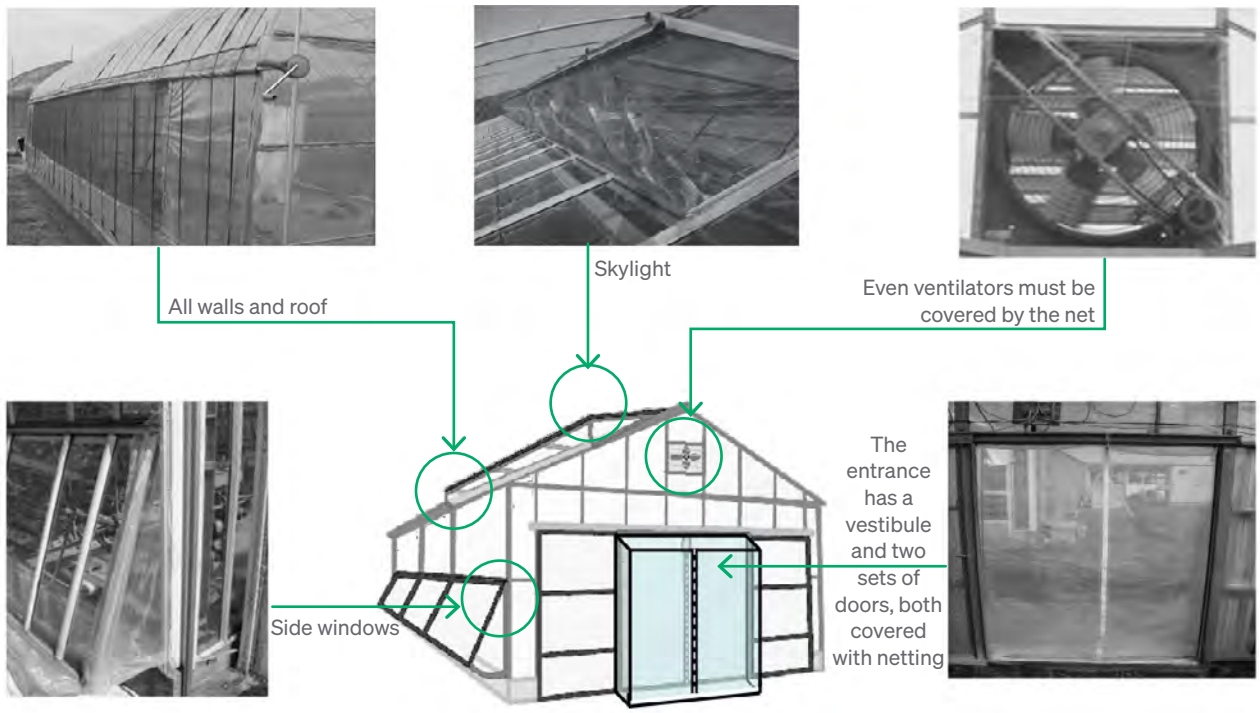

Figure 3

Covering glasshouses with nets to prevent the escape of *Bombus terrestris*: Japanese Government Accreditation Scheme [24]

DNA analysis, the NIES has already found genetic diversity among local populations of *B. ignitus*; this observation indicates the need to account for the genetic endemism of *B. ignitus* before using it in commercialised colonies [26].

Recent shortages in the supply of the European honeybee, *A. mellifera*, in Japan are increasing the demand for bumblebees as pollinators of various agricultural crops in addition to the tomato. If commercial bumblebees are used to pollinate additional crops, the ecological risks they pose can become more difficult to control.

Commercialisation of beetles as companion animals in Japan

In Japan, in particular, a culture of breeding beetles as companion animals has been deeply rooted for a long time. The Plant Protection Law of Japan, enacted in 1950, prohibited the importation of beetles, but the prohibition was lifted in 1999. A large number of exotic beetles have since been brought in and sold, while the number of foreign beetle species permitted for importation has continued to increase. Only five years after the ban was lifted, the number of foreign species in Japan exceeded 700. The size of the market was estimated to be no less than several billion yen at that time, with some estimates reaching as high as 10 billion yen [27]. Beetle imports have likely been accelerated by the global economy and liberalisation of world trade.

This unusual boom has brought with it the criminal smuggling of rare and endangered species [28]. For example, in Japan, it is possible to purchase a rare species of stag beetle, *Dorcus antaeus*, native to Bhutan, Nepal and India, although the governments of these countries have prohibited collection of any wild organisms. The price of this beetle was once as high as 600,000 Japanese yen (US$ 5,000) per specimen. As there is no law in Japan to prohibit insects from being imported from protected areas, many insect collectors and traders have gone abroad to smuggle stag beetles. One popular among Japanese collectors is *Lucanus cervus akbesianus*, a rare subspecies found only in the Nur Mountains in southern Turkey. A local environmental agency has warned that overharvesting is bringing this beetle close to extinction [29].

Large rhinoceros beetles from South America are also popular in Japan. *Dynastes hercules* and *Dynastes satanas* are being collected in countries where insect collecting is banned and exported to Japan in large numbers [30]. These beetles are endangered species in South America, especially *Dynastes satanas*, which is regulated under the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES).

Ecological risks of stag beetles as invasive alien species

Many ecologists and entomologists have warned about the ecological risks that will be caused by increased insect trade. When exotic beetles were first imported, the public and even many entomologists believed that large tropical insects could not survive the winter in Japan if they escaped into the field. But almost all stag beetles from tropical and subtropical

areas live at relatively high altitudes, where the climate is not so different from that of Japan [31]. Furthermore, almost all stag beetles spend their larval stage in decayed trees or in soil, which are more stable habitats than open air, thus allowing them to survive, even in colder regions than their original habitat. These characteristics suggest that many species of exotic stag beetles have the potential to be naturalised in Japan [31]. It is important to consider and assess ecological impacts that will be caused by their naturalisation.

The native stag beetles with an ecological niche similar to that of exotic stag beetles will suffer serious and direct impacts. The wild populations of native stag beetles are already rapidly decreasing because their habitats have been artificially disturbed. Some species are close to endangered [32]. Research by the NIES indicated that *Dorcus titanus*, a species widespread throughout the Japanese archipelago and South-East Asia, has subspecies lineages with distinct genetic differentiation in morphology and DNA sequence on each island, diverged through millions of years. Experimental crossbreeding of these subspecies has produced fertile hybrids, suggesting that the escape of foreign companion animals into the wild may lead to genetic introgression and consequent loss of genetic endemism (Figure 4) [27].

Parasite invasion from imported stag beetles

As with bumblebee imports, the exotic beetles brought into Japan may introduce new species of parasites, spreading disease among Japanese beetle populations. The parasites of stag beetles have been little investigated, and it is difficult to estimate the ecological risks of their invasion. But the NIES has studied the genetic variation and phylogeny of the endoparasitic canestriniid mites *Haitlingeria* sp. and *Sandrophela* sp. living on the surface of stag beetles [27, 33]. The analysis suggests that each of the diverse species and lineages of stag beetle has its own unique parasite (Figure 5). The translocation of beetles to different habitats may turn these parasites into invasive alien species [27, 33, K. Goka, unpublished data].

Through rearing experiments, the NIES has confirmed that the larvae of two native Japanese species, *Dorcus rectus* and *Dorcus titanus pilifer*, have the ability to fix nitrogen in the air [34, 35]. This ability is most likely due to microorganisms associated with the larvae. It seems that only adult females of the stag beetle have an organ called a mycangium and pass useful microorganisms to the next generation [36]; it is possible that the microorganisms involved in nitrogen fixation are vertically transmitted by this mechanism. Symbiotic microorganisms seem to be unique to each species and local population. Stag beetles play an important role in the material cycle of the forest, including nitrogen fixation. The risk is that the introduction and release of alien species will disturb the symbiotic relationship between stag beetles and microorganisms endemic to each region and as a result affect the material cycle system of forests.

Regulation of imported stag beetles

Out of concern about the ecological impact if exotic beetles become wild in Japan's natural environment, Japanese officials and researchers discussed what restrictions should be imposed on the importation and sale of exotic beetles. Many beetles have already been imported and are being kept in households, so it is possible that imposing sudden restrictions would lead to mass dumping of captive animals. The government of Japan, in consultation with a group of experts, decided to launch a campaign aimed at breeders to improve the ethics of breeding beetles. Its message to breeders is, 'Do not let them escape and keep them until the end of their

Figure 4

The cross between a male *Dorcus titanus castanicolor* from Japan and a female *D. t. titanus* from Indonesia (left), and an F1 hybrid male obtained from the cross (right)

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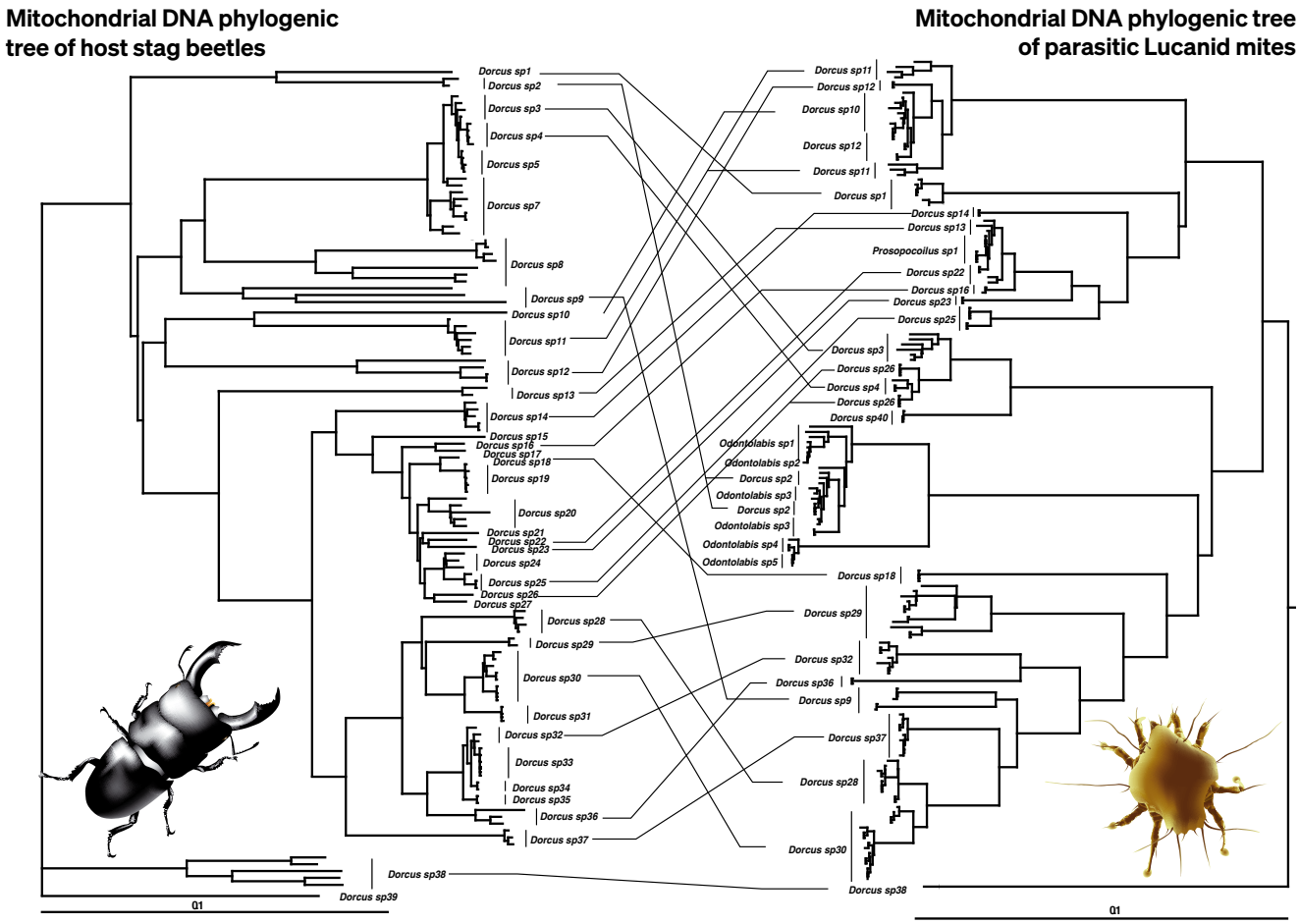

Figure 5

Phylogenetic associations between host *Dorcus* species and parasitic canestriniid mites, based on neighbour-joining trees of mitochondrial DNA CO gene sequences [27, with unpublished data]

life'. Fortunately, there have been few reported cases of exotic beetle species being naturalised in Japan.

Other concerns: tarantulas, centipedes and scorpions

Japan also imports a large number of other arthropods as companion animals. Spiders account for the largest proportion—about 90%—of arthropods intercepted by Japanese Customs [28]. Tarantulas (Figure 6), in particular, are popular as companion animals internationally [37] and are rapidly gaining popularity in Japan. Most arachnids are imported from the wild, indicating the smuggling of species classed as endangered under CITES.

Other popular companion species include centipedes and scorpions, which are caught abroad and sold on the Internet every day. Sales between individuals via the Internet make it difficult to trace imported species and detect illegal sales. The ecological information on these arthropods has not been fully investigated, and it is not yet possible to assess the impact of overexploitation on wild populations at their source or the ecological risk they pose as alien species if they are released or escape in the importing country.

Conclusions

In view of the problems posed by agricultural pollinators and companion insects in Japan as described in this paper, individuals and organisations that trade in insects for commercial use or regulate such trade need to be aware of the following risks:

- overexploitation of rare species collected from their native habitats
- trafficking of species for which collection and sale is prohibited
- establishment of escaped individuals in the new location as alien species
- introduction of associated microorganisms and parasites.

In Japan, many people have experience keeping insects, but general awareness of the problems caused by the sale and keeping of non-native insects is low [38]. Arthropods are

Figure 6

Examples of tarantula imports and sales in Japan

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important organisms that support the foundations of ecosystems, and their conservation requires greater attention and care in the future.

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La biologie de la conservation face aux échanges commerciaux d'insectes au Japon : l'exemple des bourdons utilisés en agriculture et des insectes de compagnie

K. Goka

Résumé

Le Japon importe un large éventail d'arthropodes pour un usage industriel mais aussi en tant qu'insectes de compagnie. Ces importations représentent une menace potentielle tant pour les écosystèmes locaux que pour leurs régions d'origine. S'agissant des deux insectes iconiques que sont les bourdons pollinisateurs et les coléoptères de compagnie, les importations posent de nombreux problèmes écologiques.

Les colonies de bourdons *Bombus terrestris* ont fortement contribué à la production agricole depuis leur première introduction au Japon dans les années 1990. Cependant, une fois installés et retournés progressivement à l'état sauvage, ils nuisent aux populations natives de bourdons par la compétition qu'ils exercent à leur encontre, la survenue d'hybridations spontanées et l'introduction de parasites. Ils représentent aussi une menace pour la reproduction des

plantes natives. L'utilisation de cette espèce en agriculture n'est actuellement autorisée que dans des serres fermées. Depuis 2000, les importations de coléoptères de compagnie ont prospéré et représentent un marché estimé à plusieurs milliards de yens. La popularité des espèces rares a entraîné une forte hausse des prix, un excès de captures et un essor de la contrebande depuis leurs pays d'origine. Ces espèces exotiques peuvent également devenir envahissantes si elles s'échappent dans la nature. Les importations de coléoptères ne font l'objet d'aucune restriction légale mais le gouvernement a lancé une campagne visant à améliorer les normes éthiques des élevages.
En outre, les tarentules, les myriapodes et les scorpions d'importation gagnent aussi en popularité. Ces espèces posent des menaces similaires à celles des abeilles et des coléoptères importés, mais l'échelle des importations et des élevages les concernant est difficile à déterminer avec certitude.
Les problèmes que peuvent générer les importations d'insectes au Japon sont les suivants : une surexploitation des espèces rares prélevées dans leurs habitats natifs ; le trafic illicite d'espèces dont la collecte et la vente sont interdites ; le risque que des insectes échappés dans la nature puissent se reproduire, l'espèce devenant alors envahissante ; le risque d'introduction de microorganismes et de parasites exotiques.

Mots-clés
Animal de compagnie – *Bombus terrestris* – Bourdon – Coléoptère – Espèce exotique – Espèce menacée d'extinction – Pollinisateur – Risque écologique.

Biología de la conservación aplicada al transporte comercial de insectos en el Japón: el ejemplo de los abejorros de uso agrícola y el de los insectos-mascota

K. Goka

Resumen
El Japón importa muy diversos artrópodos que son destinados a usos industriales o utilizados como mascota. Estas importaciones pueden entrañar una amenaza para los ecosistemas, tanto en los lugares de destino como en sus regiones de origen. Dos importaciones emblemáticas de insectos que causan problemas ecológicos son las de abejorros de uso agrícola y las de escarabajos utilizados como mascota.
Desde que en el decenio de 1990 fueron introducidas por primera vez en el Japón, las colonias de abejorro común (*Bombus terrestris*) han revestido gran utilidad para la producción agrícola. Ahora bien, su progresiva feralización resulta perjudicial para las poblaciones de abejorros nativos, debido a los fenómenos de competencia, hibridación e introducción de parásitos que la acompañan, y pone también en peligro la reproducción de ciertas plantas nativas. Actualmente solo está autorizado el uso agrícola de esta especie en invernaderos cerrados con malla.
Desde el año 2000 vienen aumentando las importaciones de escarabajos que se utilizan como mascota, hasta el punto de que ello supone hoy, según los cálculos, un mercado de miles de millones de yenes. La popularidad de ciertas especies raras ha traído consigo una fuerte subida de los precios, capturas excesivas y contrabando de esas especies desde sus países de origen. Si escapan al medio natural, además, estas especies exóticas también pueden devenir invasoras. Las importaciones de escarabajos no están sujetas a restricciones claras, aunque el Gobierno tiene instituida una campaña para mejorar la deontología de las normas de cría.
Por otro lado, cada vez hay más querencia por tarántulas, ciempiés y escorpiones importados. Estas especies entrañan amenazas análogas a las que plantean escarabajos y abejas de importación, pero resulta difícil determinar cuál es la situación actual por lo que respecta a su importación y cría.
La importación de insectos al Japón puede acarrear los siguientes problemas: la sobreexplotación de especies raras en su hábitat nativo; el tráfico de especies cuya captura y venta están prohibidas; el riesgo de que haya ejemplares que escapen y se reproduzcan hasta devenir especie invasora; y la introducción de microorganismos y parásitos foráneos.

Palabras clave
Abejorro – *Bombus terrestris* – Escarabajo – Especie alóctona – Especie en peligro – Mascota – Polinizador – Riesgo ecológico.

References

[1] Mumford J.D. & Quinlan M.M. (2022). – Introduction. *In* Safety, regulatory and environmental issues related to international trade of live insects (J.D. Mumford & M.M. Quinlan, eds). *Rev Sci Tech,* **41** (1), 15–18. https://doi.org/10.20506/rst.41.1.3301

[2] Lautenbach S., Seppelt R., Liebscher J. & Dormann C.F. (2012). – Spatial and temporal trends of global pollination benefit. *PLoS ONE*, **7** (4), e35954. https://doi.org/10.1371/journal.pone.0035954

[3] Velthuis H.H.W. & van Doorn A. (2006). – A century of advances in bumblebee domestication and the economic and environmental aspects of its commercialization for pollination. *Apidol.*, **37** (4), 421–451. https://doi.org/10.1051/apido:2006019

[4] Goulson D. (2003). – Bumblebees: their behaviour and ecology. Oxford University Press, New York, United States of America, 235 pp.

[5] Kunitake K.Y. & Goka K. (2006). – A new aspect of the alien species problem: ecological risks caused by commercialization of the native bumblebee, *Bombus ignitus* [in Japanese]. *Nat. Insects*, **41** (4), 37–40.

[6] Ministry of Agriculture, Forestry and Fisheries (MAFF) (Japan) (2020). – Introduction to bumblebees [in Japanese]. MAFF, Tokyo, Japan, 14 pp. Available at: https://www.maff.go.jp/j/chikusan/kikaku/lin/sonota/attach/pdf/bee-15.pdf (accessed on 6 February 2022).

[7] Washitani I. (1998). – Conservation–ecological issues of the recent invasion of *Bombus terrestris* into Japan [in Japanese]. *Jpn. J. Ecol.*, **48** (1), 73–78.

[8] Matsumura C., Yokoyama J. & Washitani I. (2004). – Invasion status and potential ecological impacts of an invasive alien bumblebee, *Bombus terrestris* L. (Hymenoptera: Apidae) naturalized in Southern Hokkaido, Japan. *Glob. Environ. Res.*, **8** (1), 51–66. Available at: https://www.researchgate.net/publication/228890812_Invasion_Status_and_Potential_Ecological_Impacts_of_an_Invasive_Alien_Bumblebee_Bombus_terrestris_L_Hymenoptera_Apidae_Naturalized_in_Southern_Hokkaido_Japan (accessed on 17 January 2022).

[9] Goulson D. (2010). – Impacts of non-native bumblebees in Western Europe and North America. *Appl. Entomol. Zool.*, **45** (1), 7–12. https://doi.org/10.1303/aez.2010.7

[10] Cresswell J.E. (2010). – Theoretical analyses of the impact of change in the composition of the pollinator fauna on a pollination system. *Appl. Entomol. Zool.*, **45** (1), 13–20. https://doi.org/10.1303/aez.2010.13

[11] Goka K. (1998). – Influence of invasive species on native species – will the European bumble-bee, *Bombus terrestris*, bring genetic pollution into the Japanese native species? [in Japanese]. *Bull. Biogeogr. Soc. Jpn.*, **53**, 91–101.

[12] Goka K. (2003). – The ecological problem caused by commercialization of the bumblebee – the past and the future [in Japanese]. *Jpn. J. Plant Protect.*, **57**, 452–456.

[13] Inoue M.N., Yokoyama J. & Washitani I. (2008). – Displacement of Japanese native bumblebees by the recently introduced *Bombus terrestris* (L.) (Hymenoptera: Apidae). *J. Insect Conserv.*, **12** (2), 135–146. https://doi.org/10.1007/s10841-007-9071-z

[14] Inoue M.N. & Yokoyama J. (2010). – Competition for flower resources and nest sites between *Bombus terrestris* (L.) and Japanese native bumblebees. *Appl. Entomol. Zool.*, **45** (1), 29–35. https://doi.org/10.1303/aez.2010.29

[15] Yokoyama J. & Inoue M.N. (2010). – Status of the invasion and range expansion of an introduced bumblebee, *Bombus terrestris* (L.), in Japan. *Appl. Entomol. Zool.*, **45** (1), 21–27. https://doi.org/10.1303/aez.2010.21

[16] Dohzono I., Kunitake Y.K., Yokoyama J. & Goka K. (2008). – Alien bumblebee affects native plant reproduction through interactions with native bumblebees. *Ecol.*, **89** (11), 3082–3092. https://doi.org/10.1890/07-1491.1

[17] Kanbe Y., Okada I., Yoneda M., Goka K. & Tsuchida K. (2008). – Interspecific mating of the introduced bumblebee *Bombus terrestris* and the native Japanese bumblebee *Bombus hypocrita sapporoensis* results in inviable hybrids. *Naturwissenschaften*, **95** (10), 1003–1008. https://doi.org/10.1007/s00114-008-0415-7

[18] Kondo I.N., Yamanaka D., Kanbe Y., Kunitake K.Y., Yoneda M., Tsuchida K. & Goka K. (2009). – Reproductive disturbance of Japanese bumblebees by the introduced European bumblebee *Bombus terrestris*. *Naturwissenschaften*, **96** (4), 467–475. https://doi.org/10.1007/s00114-008-0495-4

[19] Oudemans A.C. (1904). – On a new genus and species of parasitic Acari. *Notes from the Leyden Museum*, **24**, 216–222. Available at: https://www.biodiversitylibrary.org/partpdf/22953 (accessed on 17 January 2022).

[20] Schmid-Hempel P. (1999). – Parasites in social insects. Princeton University Press, Princeton, United States of America, 409 pp.

[21] Goka K., Okabe K., Niwa S. & Yoneda M. (2000). – Parasitic mite infestation in introduced colonies of European bumblebees, *Bombus terrestris* [in Japanese with English summary]. *Jpn. J. Appl. Entomol. Zool.*, **44** (1), 47–50. https://doi.org/10.1303/jjaez.2000.47

[22] Goka K., Okabe K. & Yoneda M. (2006). – Worldwide migration of parasitic mites as a result of bumblebee commercialization. *Popul. Ecol.*, **48** (4), 285–291. https://doi.org/10.1007/s10144-006-0010-8

[23] Goka K., Okabe K. & Yoneda M. (2007). – Bumblebee commercialization has caused worldwide migration of parasitic mites. *In* Acarology XI (J.B. Morales-Malacara, V. Behan-Pelletier, E. Ueckermann, T.M. Pérez, E.G. Estrada-Venegas & M. Badii, eds). Universidad Nacional Autonoma de Mexico, Mexico City, Mexico, 25–32.

[24] Goka K. (2010). – Introduction to the special feature for ecological risk assessment of introduced bumblebees; status of the European bumblebee, *Bombus terrestris*, in Japan as a beneficial pollinator and an invasive alien species. *Appl. Entomol. Zool.*, **45** (1), 1–6. https://doi.org/10.1303/aez.2010.1

[25] Kishi S. & Goka K. (2017). – Review of the invasive yellow-legged hornet, *Vespa velutina nigrithorax* (Hymenoptera: Vespidae), in Japan and its possible chemical control. *Appl. Entomol. Zool.*, **52** (3), 361–368. https://doi.org/10.1007/s13355-017-0506-z

[26] Tokoro S., Yoneda M., Kunitake K.K. & Goka K. (2010). – Geographic variation of mitochondrial DNA in *Bombus ignitus* (Hymenoptera: Apidae). *Appl. Entomol. Zool.*, **45** (1), 77–87. https://doi.org/10.1303/aez.2010.77

[27] Goka K., Kojima H. & Okabe K. (2004). – Biological invasion caused by commercialization of stag beetles in Japan. *Glob. Environ. Res.*, **8** (1), 67–74. Available at: http://www.airies.or.jp/journal_08-1eng.html (accessed on 17 January 2022).

[28] Kitade T. & Naruse Y. (2020). – Crossing the red line: Japan's exotic pet trade. TRAFFIC, Japan Office, Tokyo, Japan. Available at: https://

www.traffic.org/site/assets/files/12923/crossing_the_red_line_en-forweb.pdf (accessed on 16 December 2021).

[29] Holden C. (2007). – Beetle battles. *Science*, **318** (5847), 25. https://doi.org/10.1126/science.318.5847.25a

[30] Berton E.F. (2020). – Why rare beetles are being smuggled to Japan at an alarming rate: big-horned rhinoceros beetles, taken from Bolivia, are ending up in Japan's illegal pet trade – and in beetle wrestling matches. *National Geographic*. Available at: https://www.nationalgeographic.com/animals/article/bolivian-beetles-smuggled-to-japan-for-fighting (accessed on 16 December 2021).

[31] Araya K. (2002). – A threat of exotic lucanid beetles to domestic species [in Japanese]. *Nat. Insects*, **37** (3), 4–7.

[32] Kojima H. (2003). – The reason which *Dorcus titanus pilifer* of Kanto area enlarges. Three hypotheses why read and solve change of stag beetles by global warming [in Japanese]. *Nat. Insects*, **38** (3), 13–19.

[33] Okabe K. & Goka K. (2008). – Potential impacts on Japanese fauna of canestriniid mites (Acari: Astigmata) accidentally introduced with pet lucanid beetles from Southeast Asia. *Biodivers. Conserv*., **17** (1), 71–81. https://doi.org/10.1007/s10531-007-9231-1

[34] Kuranouchi T., Nakamura T., Shimamura S., Kojima H., Goka K., Okabe K. & Mochizuki A. (2006). – Nitrogen fixation in the stag beetle, *Dorcus* (*Macrodorcus*) *rectus* (Motschulsky) (Col., Lucanidae). *Appl. Entomol. Zool.*, **130** (9–10), 471–472. https://doi.org/10.1111/j.1439-0418.2006.01080.x

[35] Kuranouchi T., Mochizuki J., Kojima H. & Goka K. (2011). – Change of the amounts of nitrogen and carbon in the decayed wood chips by rearing of the stag beetle, *Dorcus titanus pilifer* (Coleoptera: Lucanidae) [in Japanese]. *Jap. J. Entomol.*, **14** (4), 276–280. https://doi.org/10.20848/kontyu.14.4_276

[36] Tanahashi M., Kubota K., Matsushita N. & Togashi K. (2010). – Discovery of mycangium and the associated xylose-fermenting yeasts in stag beetles (Coleoptera: Lucanidae). *Naturwissenschaften*, **97** (3), 311–317. https://doi.org/10.1007/s00114-009-0643-5

[37] Fukushima C., Mendoza J.I., West R.C., Longhorn S.J., Rivera E., Cooper E.W.T., Hénaut Y., Henriques S. & Cardoso P. (2019). – Species conservation profiles of tarantula spiders (Araneae, Theraphosidae) listed on CITES. *Biodiversity Data J.*, **7**, e39342. https://doi.org/10.3897/BDJ.7.e39342

[38] Iwanishi S. & Sawahata T. (2015). – The actual condition of breeding and awareness of issues of alien species rhinoceros and stag beetles by museum visitors: the findings from a questionnaire survey in a museum of natural science. *Jap. J. Environ. Educ*., **25** (1), 168–175. https://doi.org/10.5647/jsoee.25.1_168

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